Growing up in a compound for doctors due to my dad's job, and with friends studying medicine, I developed an interest in healthcare early on. The first time I worked on the healthtech intersection was after studying ML in my bachelor's. My mind went to how biases in healthcare literature would be cemented in the upcoming age of AI healthcare. I compiled my findings into a blog and messaged researchers working on such models at the time, proposing methods to check fairness. Received a few positive responses, but a lack of resources prevented any actual work. That's what motivated me to pursue a master's and be part of an institute again. Satisfied with my ML knowledge base, I focused on applying it. I took a health innovations course covering the EU healthtech landscape and the end-to-end process of bringing innovations through it. A hospital project course investigating EHR interoperability challenges in Sweden, learning the gap between implemented and existing standards. A small consultancy project forecasting patient arrivals at an emergency department. Most recently, I've started a pilot with a hospital in Saudi Arabia on using AI to read electrophoresis results and speed up lab turnaround times. None are deep research projects individually, but the breadth of exposure means I understand the structures and common problems. A PhD coming from that awareness is more likely to produce something workers actually want to use, and that fits the system.

What draws me to this position is the data problem more than the disease. Wearables and video analytics are inherently messy: variable recording conditions, missing segments, individual differences in movement patterns that aren't noise but signal. Deriving patient-specific biomechanical parameters from that kind of data, rather than from controlled lab measurements, is the kind of problem where good ML design matters as much as model choice. My work so far has mostly involved text and tabular data, but the underlying challenge is familiar: real-world data collected without a controlled protocol, where the pipeline design and data construction decisions shape whether the model output means anything.

The pilot in Saudi Arabia is where I've thought most seriously about what good healthcare AI design actually requires. The lab data is heavily imbalanced, and standard metrics would give false confidence. More useful is mapping the physician's actual workflow and asking whether the AI addition is a clear improvement, or just a lateral move. Second-order effects matter too. A system that handles routine cases well can quietly change the difficulty distribution of what clinicians see, in ways that affect adoption without showing up in any evaluation metric. My biggest motivation for a PhD position is knowing the research is designed with those realities in mind and has a realistic path to being used.

I don't have a background in biomechanics or musculoskeletal modelling. I have no OpenSim experience. What I bring is experience with multimodal data pipelines, working under label uncertainty, and the habit of treating data construction as a first-class research problem rather than a preprocessing step. The domain knowledge is something I'd expect to build through the coursework and the team. The earliest I can start is August 2026.